EXT OUT OF AN M-PSEUDO-COHERENT COMPLEX COMMUTES WITH FILTERED COLIMITS BELOW DEGREE MINUS M

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Ext out of a complex with vanishing cohomology above m vanishes in degrees at most minus m. Derived Categories, Lemma \ref{derived-lemma-negative-exts}, cited at proof lines 16609-16610. Makes the higher terms of the long exact sequence disappear.
- Ext groups of a bounded complex of finite free modules are computed by the Hom complex. Derived Categories, Lemma \ref{derived-lemma-morphisms-from-projective-complex}, cited at proof lines 16617-16618. Reduces the remaining statement to a statement about Hom complexes.
- Filtered colimits of modules are exact. Algebra, Lemma \ref{algebra-lemma-directed-colimit-exact}, cited at proof lines 16623-16624. Passes the colimit through the cohomology of the Hom complex.

Source reference. Original lemma 0G8W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20281–20281. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 06XS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8861–8861. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7860–7860. Navigation pointer to a statement the

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Source reference. Original lemma 00DB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 491–491. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $M = \operatorname{colim} M_i$ be a filtered colimit of R-modules. Let $K \in D(R)$ be m-pseudo-coherent. Then $\operatorname{colim} \operatorname{Ext}^n_R(K, M_i) = \operatorname{Ext}^n_R(K, M)$ for $n < -m$ and $\operatorname{colim} \operatorname{Ext}^{-m}_R(K, M_i) \to \operatorname{Ext}^{-m}_R(K, M)$ is injective.*

Proof. By definition we can find a distinguished triangle

$$E \to K \to L \to E[1]$$

in $D(R)$ such that E is represented by a bounded complex of finite free R-modules and such that $H^i(L) = 0$ for $i \geq m$. Then $\operatorname{Ext}^n_R(L, N) = 0$ for any R-module N and $n \leq -m$, see Derived Categories, Lemma 06XS. By the long exact sequence of Ext associated to the distinguished triangle we see that $\operatorname{Ext}^n_R(K, N) \to \operatorname{Ext}^n_R(E, N)$ is an isomorphism for $n < -m$ and injective for $n = -m$. Thus it suffices to prove that $M \mapsto \operatorname{Ext}^n_R(E, M)$ commutes with filtered colimits when E can be represented by a bounded complex of finite free R-modules $E^\bullet$. The modules $\operatorname{Ext}^n_R(E, M)$ are computed by the complex $\operatorname{Hom}_R(E^\bullet, M)$, see Derived Categories, Lemma 064B. The functor $M \mapsto \operatorname{Hom}_R(E^p, M)$ commutes with filtered colimits as E^p is finite free. Thus $\operatorname{Hom}_R(E^\bullet, M) = \operatorname{colim} \operatorname{Hom}_R(E^\bullet, M_i)$ as complexes. Since filtered colimits are exact (Algebra, Lemma 00DB) we conclude. □